Discrete Math 2 - Semi Summary

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1.1 Relations

What is a Relation?

A relation is just a set of ordered pairs. For example:

$$R = \{(a, b), (c, d), ...\}$$

The first element of each pair comes from set A. The second element of each pair comes from set B (or A if it's a relation within the same set). We say a is "related to" b if $(a, b) \in R$.

"Related" depends on how the relation is defined. For example:

1. Equality Relation $(=)$: a is related to b if $a = b$.
   - Example: $(2, 2) \in R$, so 2 is related to itself.
2. Divides Relation $(a|b)$: a is related to b if a divides b (i.e., $b = ka$ for some integer k).
   - Example: $(2, 6) \in R$, so 2 is related to 6 because $2|6$.
3. Custom Relation $(a, b) \doteq (c, d)$ means $a \cdot b = c \cdot d$.
   - Example: $(2, 3) \doteq (1, 6)$, so $(2, 3)$ is related to $(1, 6)$ because $2 \cdot 3 = 1 \cdot 6$.

If (a, b) is in a relation R:

- a is "related to" b under the rules of R.
- The meaning of "related" depends entirely on how R is defined.

1.1.1 Equivalence relations

An equivalence relation:

- Groups elements into "equivalent" classes.
- Follows reflexivity, symmetry, and transitivity.
- Examples include equality, modular arithmetic, and geometric similarity.

1.1.2 Reflexive relations

A binary Relation R defined on a Set A is said to be reflexive if for each element a ∈ A, we have aRa. R is reflexive if for each a ∈ A, (a,a) ∈ R.

Lets say we have Set

$$A = \{1, 2, 3\}$$

If R contains sets that state that all elements in A relate to themselves, then R is reflexive. This means that R must contain $(1, 1)(2, 2)$ and $(3, 3)$. If the answer contains $(1, 2)$ or $(2, 3)$ etc. it doesn't affect the answer, it only describes the connection between elements.

$$R = \{(0, 0), (1, 2), (1, 1), (2, 2)(3, 3)\}$$

The formula for reflexive relation states the number of reflexive relations in a given set.

The formula is:

$$N = 2 \cdot n(n - 1)$$

where

N = Number of reflexive relations

n = the number of elements in the set

1.1.3 Irreflexive/reflexive relations

In set A, a relation R is anti-reflexive (or irreflexive) if no element of the set is related to itself. Thus, R is anti-reflexive if $(a, a) \in R$ for all A.

A relation R is irreflexive if:

$$(a, a) \notin \text{ for all } a \in A.$$

In simple terms:

- No element of the set A is related to itself.
- This means $(a, a) \in R$ must never happen for any $a \in A$.

1.1.4 Symmetric relations

Symmetry

Symmetry means relationships go both ways. If x relates to y, then y must also relate to x.

A relation R on a set A is symmetric if:

1. Whenever $(x, y) \in R$, then (y, x) must also be in R.
2. In simpler terms: If x is related to y, then y must also be related t

Let say we have Set

$$A = \{1, 4, 5, 7\}$$

And the relation R is:

$$R = \{(1, 5), (5, 1), (4, 7), (7, 4)\}$$

Look at the pairs in R:

- It has (1,5), and it also has the "flipped" pair (5,1).
- It has (4,7), and it also has the "flipped" pair (7,4).

Since every pair (x,y) has its corresponding pair (y,x), the relation R is symmetric.

Even if a single ordered pair fails to meet the condition for the symmetric relation, the relation is considered not symmetric.

The “is equal to” relation is a symmetric relation since m = n, implies that n = m. The relation “is parallel to” defined on the set of straight lines is a symmetric relation since if a line l is parallel to m, we know that the line m is also parallel to the line l.

Number of Symmetric Relations

We can find the total number of symmetric relations on a given set with n elements. Let us consider a set A with n elements on which a relation R is defined. The total number of symmetric relations on A is given by $2^{\frac{n(n+1)}{2}}$.

Symmetric Relation Formula

Number of symmetric relations on a set with n elements = $N = 2^{\frac{n(n+1)}{2}}$ where n is the number of elements in the set.

Asymmetry

Asymmetry is the opposite of a symmetric relation, meaning A relation R is asymmetric if:

$$(a, b) \in R \implies (b, a) \notin R$$

In simpler terms:

- If a is related to b, then b cannot be related to a.
- Both direction ((a, b) and (b, a)) can never exist together in R.

Antisymmetry

Antisymmetry means (a, b) and (b, a) can only both exist if $a = b$.

A relation R is antisymmetric if:

$$(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b$$

In simpler terms, if a is related to b and b is related to a, then a and b must be the same.

1.1.5 Transitive Relations

A relation is said to be a transitive relation, if "a is related to b" and "b is related to c" implies that "a is related to c."

A binary relation R defined on set A is considered a transitive relation, if for Set $A = a, b, c$, $(a, b) \in R$, and $(b, c) \in R$ implies that ($\implies$) $(a, c) \in R$.

1.2 Statements

A statement (or proposition) is a declarative sentence that is either true or false, but not both. If a sentence does not have truth values (true or false), they are not regarded as statements. The sentences below are considered to be *primitive* statements since they can't be broken down any simpler.

E.g.

- p: Combinatorics is a required course for sophomores.
- q: Margaret Mitchell wrote *Gone with the Wind*.
- r: $2 + 3 = 5$

There are two ways to obtain new statements from existing ones:

1. Transform p to *neg* p, which indicates the negation and is read not p. So the statement p becomes $\neg$ p: "Combinatorics is not a required course for sophomores." The negation of a primitive statement is not considered to be a primitive statement.
2. Combine two or more statement into a compound statement using the following *logical connectives*
   - **Conjunction:** The conjunction of the statements p, q is denoted by p $\wedge$ q, which is read "p and q". So the statement becomes "Combinatorics is a required course for sophomores, **and** Margaret Mitchell wrote *Gone with the Wind*."
   - **Disjunction:** The expression p $\vee$ q denotes the disjunction of the statements p, q and is read "p and q". So the statement becomes "Combinatorics is a required course for sophomores, **or** Margaret Mitchell wrote *Gone with the Wind*." There are two types of or:
     (a) **Inclusive or**: Compound statement p $\vee$ q is true if one or the other of p, q is true or if both of the statements p, q are true.
     (b) **Exclusive or**: Compound statement p $\veebar$ q is true if one or the other of p, q is true but not both of the statements p, q are true.
   - **Implication:**

Table 2.1

p	$\neg p$
0	1
1	0

Table 2.2

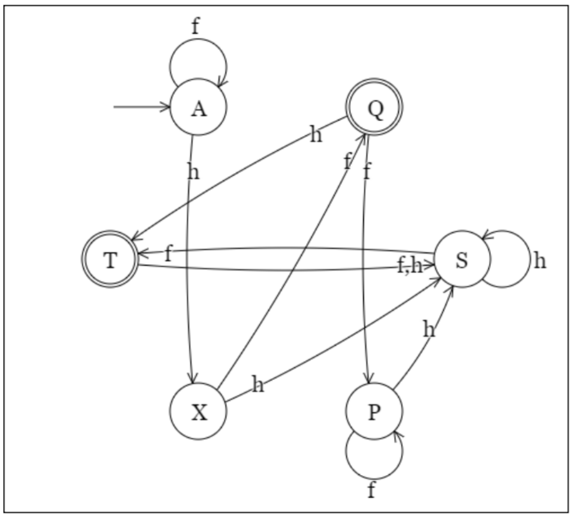
p	q	$p \wedge q$	$p \vee q$	$p \veebar q$	$p \rightarrow q$	$p \leftrightarrow q$
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	0
1	1	1	1	0	1	1

2 Delprøve 2

2.1 Oppgave 1

2.1.1 Minimize the DFA

Minimize the following DFA.

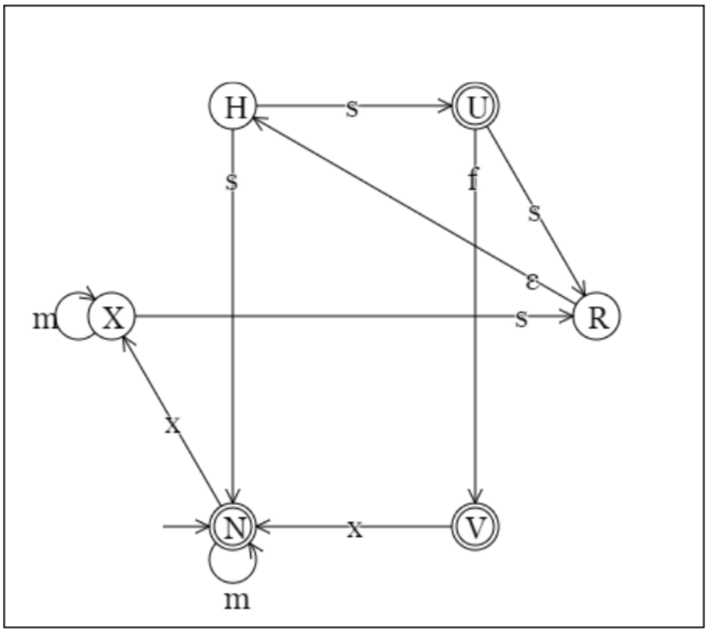


Write the answer as a set of sets of equivalent states.

Figure 1: Minimize the following DFA. Write the answer as a set of sets of equivalent states.

2.1.2 Translate the DFA

Translate the following DFA to a right-regular grammar.



Write the answer as a 4-tuple, separating the rule per non-terminal. Use the names of the states for their respective nonterminals.

Figure 2: Translate the following DFA to a right-regular grammar. Write the answer as a 4-tuple, separating the rule per non-terminal. Use the names of the states for their respective nonterminals

2.2 Oppgave 2

2.2.1 Expand Assert

Apply the refinement rule "Expand Assert" (see below) to the following specification statement.

Specification statement: { c < 8 and c = b } b,t,c:=b',t',c' | b' < b

Refinement rule Expand Assert (Exp)

{pre1 and pre2}

[=

{pre1} ; {pre2}

Fill in the correctly refined specification statement. You can copy parts of the original statement.

Figure 3: Apply the refinement rule "Expand Assert" (see figure) to the following specification statement. Fill in the correctly refined specification statement. You can copy parts of the original statement.

2.2.2 Introduce constant k

Apply the refinement rule "Introduce Constant" (see below) to the following specification statement in order to introduce a constant K of type Nat with the value 8.

Specification statement: { 17 <= m } k,m:=k',m' | m' <= 19

Refinement rule Introduce Constant (InC)

```
   {pre} v,x:=v',x' | rel
[=
   CON C:T | C=x . {C=x and pre} v,x:=v',x' | rel
```

Fill in the correctly refined specification statement. You can copy parts of the original statement.

Figure 4: Apply the refinement rule "Introduce Constant" (see figure) to the following specification statement in order to introduce a constant K of type Nat with the value 8.

2.2.3 Introduce constant T

Apply the refinement rule "Introduce Constant" (see below) to the following specification statement in order to introduce a constant T of type Nat with the value 8.

Specification statement: { 18 < c } c,x,h:=c',x',h' | c' != c

Refinement rule Introduce Constant (InC)

```
   {pre} v,x:=v',x' | rel
[=
   CON C:T | C=x . {C=x and pre} v,x:=v',x' | rel
```

Fill in the correctly refined specification statement. You can copy parts of the original statement.

Figure 5: Apply the refinement rule "Introduce Constant" (see below) to the following specification statement in order to introduce a constant T of type Nat with the value 8.

2.2.4 Introduce local variable

Apply the refinement rule "Introduce Local Variable" (see below) to the following specification statement in order to introduce a variable R of type Nat with initial value 2.

Specification statement: { 9 > o or o <= 17 } o,r:=o',r' | r' != o and o' > r

Refinement rule Introduce Local Variable (InV)

```
   {pre} v:=v' | rel
[=
   VAR l:T . l:=E; {pre} v,l:=v',l' | rel
```

Fill in the correctly refined specification statement. You can copy parts of the original statement.

Figure 6: Apply the refinement rule "Introduce Local Variable" (see below) to the following specification statement in order to introduce a variable R of type Nat with initial value 2.

2.2.5 Expand Assert 2

Apply the refinement rule "Expand Assert" (see below) to the following specification statement.

Specification statement: { x != u and g = x } u,x,g:=u',x',g' | g' > x => u' = g

Refinement rule Expand Assert (Exp)

{pre1 and pre2}

[=

{pre1} ; {pre2}

Fill in the correctly refined specification statement. You can copy parts of the original statement.

Figure 7: Apply the refinement rule "Expand Assert" (see below) to the following specification statement.

2.3 Oppgave 3

Consider the following relation R over integers.

xRy iff x + y = 54

Check the properties of the relation R.

The relation is reflexive:	
The relation is symmetric:	
The grammar is antisymmetric:	
The grammar is transitive:	

Figure 8: Consider the following relation R over integers. Check the properties of the relation R.

2.4 Oppgave 4

2.4.1

Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.

positive cases: xfffqa, xqa, xxfffffqa, xxffq, xxfqa, xxxfffq, xxxfq, xxxxffqa, xxxxxfffq, xxxxxq

negative cases: fffqaa, q, x, xfffqaa, xfffqq, xffqqq, xqaa, xqqq, xxfffqaaa, xxxfffqaaa

You must fill in all four parts of the answer to be graded.

x+	f*	q	a?

Figure 9: Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.

2.4.2

Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.
positive cases: zah, zahhh, zahhhhh, zzah, zzahhh, zzza, zzzahhh, zzzahhhh, zzzzahh, zzzzzahhh
negative cases: a, aaahhhppp, hhhpp, z, zaaa, zap, zz, zzaaa, zzhhh, zzzhhpp

You must fill in all four parts of the answer to be graded.

z+	a?	h*	p?

Figure 10: Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.

2.4.3

Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.
positive cases: hhhhhukk, hhhhhukkk, hhhhukkk, hhhuk, hhuk, hukk, hukkkk, ukk, ukkk, ukkkkk
negative cases: hhhuukkx, hhuuukx, huukkx, huuukkxx, k, u, ukx, uuk, uuukk, uxx

You must fill in all four parts of the answer to be graded.

h+	u?	k+	x

Figure 11: Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.

2.5 Oppgave 5

Compute the equivalence classes of the set S below with respect to the equivalence relation $=\%_5=$ (same remainder in division by 5).

S = $\{-19, -18, -16, -15, -12, -8, -7, -3, 0, 1, 3, 5, 6, 8, 9, 11, 12, 15, 18, 20\}$

The result must be given as a set of sets.

Figure 12: Compute the equivalence classes of the set S below with respect to the equivalence relation $=\% =$ (same remainder in division by 5)

2.6 Oppgave 6

2.6.1

Given the set $[1, 2, 3, 4, 5]$, and relations $\{[1, 1], [4, 1], [3, 5], [5, 1], [3, 2], [5, 5]\}$.

1. Fill in the relations matrix below such that all given relations are marked by a "**1**" in the correct position.
2. Then extend the set of relations by adding mandatory "**1**"'s so that the final set of relations is an equivalence relation.
3. All other entries in the matrix must be marked "**0**".

-	1	2	3	4	5
1	1	0	0	0	1
2	0	0	1	0	0
3	0	0	0	0	0
4	0	1	0	0	0
5	0	0	1	0	1

Figure 13: Given the set and relations.

2.6.2

Given the set $[1, 2, 3, 4, 5]$, and relations $\{[2, 4], [3, 3], [5, 2], [5, 5]\}$.

1. Fill in the relations matrix below such that all given relations are marked by a "**1**" in the correct position.
2. Then extend the set of relations by adding mandatory "**1**"'s so that the final set of relations is an equivalence relation.
3. All other entries in the matrix must be marked "**0**".

$$\begin{array}{c|c} - & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

Figure 14: Given the set and relations.

2.7 Oppgave 7

Let R be a relation over $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

We know that R is an equivalence relation and a partial order.

Moreover, $(6, 8)$ is not in R.

Select all pairs that are in R.

- ☐ $[1, 4]$
- ☐ $[2, 5]$
- ☐ $[3, 9]$
- ☐ $[4, 6]$
- ☐ $[5, 9]$
- ☑ $[6, 6]$
- ☐ $[6, 8]$
- ☐ $[7, 6]$
- ☐ $[7, 9]$
- ☐ $[8, 3]$
- ☐ $[8, 6]$
- ☑ $[8, 8]$
- ☐ $[9, 3]$
- ☐ None of the above

Figure 15: Let R be a relation over.

2.8 Oppgave 8

Sort the following conditions according to strength.

$x > 3$

$x^2 > 3^2$

$5x = 30$

$x > 4$

$x = x$

Hints:

- **A** is stronger than **B** if **A** implies **B**.

strongest:	
strong:	
medium:	
weak:	
weakest:	

Figure 16: Sort the following conditions according to strength.

2.9 Oppgave 9

2.9.1

Consider the following regular expression.

ex = y[A-J]+

Which of the following regular expressions accepts at least the same words as ex?

- [ ] y?[A-J]?
- [x] y?[A-J]*
- [ ] y?[A-J]+
- [ ] y*[A-J]?
- [ ] y*[A-J]*
- [x] y*[A-J]+
- [ ] y+[A-J]?
- [x] y+[A-J]*
- [ ] y+[A-J]+
- [ ] None of the above

Figure 17: Consider the following regular expression.

2.9.2

Consider the following regular expression.

ex = [A-J]??[0-9]+

Which of the following regular expressions accepts at least the same words as ex?

- [ ] [A-J]?[0-9]?
- [x] [A-J]?[0-9]*
- [x] [A-J]?[0-9]+
- [ ] [A-J]*[0-9]?
- [x] [A-J]*[0-9]*
- [x] [A-J]*[0-9]+
- [ ] [A-J]+[0-9]?
- [ ] [A-J]+[0-9]*
- [ ] [A-J]+[0-9]+
- [ ] None of the above

Figure 18: Consider the following regular expression.

2.9.3

Consider the following regular expression.

ex = X??[A-J]+

Which of the following regular expressions accepts at least the same words as ex?

- [ ] X?[A-J]?
- [x] X?[A-J]*
- [x] X?[A-J]+
- [ ] X*[A-J]?
- [x] X*[A-J]*
- [x] X*[A-J]+
- [ ] X+[A-J]?
- [ ] X+[A-J]*
- [ ] X+[A-J]+
- [ ] None of the above

Figure 19: Consider the following regular expression.

2.10 Oppgave 10

2.10.1

Which temporal logic statement matches the English sentence "Some darkness leads to a later light."?

Figure 20: Which temporal logic statement matches the English sentence "Some darkness leads to a later light."?

3 Handin 4: Languages, Machines, Regular expressions

3.1 Task 1, Regex

Write a regular expression that matches the following positive cases and does not match the following negative cases. Try to make the regular expression as general as possible.
Hint: the strings should denote decimal numbers, binary numbers (with the b), and hexadecimal numbers (with the x).
positive: 0, 10, 31, 1b, 0B, 12x, 2aX, 1234567890, 11111000b, 123456789abcdefX, -0, -1b, -1bx
negative: 00, 010, 31b, 1c, 0d, xx, 0cX, a, 2b, 3B, gX, -a, –1, 1-1

To solve this problem, you need to create a regular expression that recognizes decimal, binary, and hexadecimal numbers while distinguishing them from invalid strings. Let's break down the pattern needed:

Decimals:

- The positive cases for decimals are plain numbers like 0, 10, 31, and so on.
- These can be optionally preceded by a negative sign (-), and they cannot start with leading zeros unless the number is zero itself (0 is valid, but 00 or 010 is not).
- Decimals are made up of digits 0-9.

Binary Numbers:

- Binary numbers are denoted with a b or B suffix after a valid sequence of binary digits (0 or 1).
- The valid binary examples include 1b, 11111000b, -1b.
- Invalid binary cases might involve using digits other than 0 and 1, such as 2b or 3B.

Hexadecimal numbers:

- Hexadecimal numbers are denoted with a suffix x or X after a sequence of valid hexadecimal digits (0-9, a-f, or A-F).
- Examples of valid hexadecimal cases are 12x, 123456789abcdefX, and -2aX.
- Invalid cases could include incorrect letters like 1c or gX.

Key Considerations:

- You need to account for an optional negative sign at the start (-).
- No leading zeros unless the number is zero itself.
- Correct suffixes for binary (b, B) and hexadecimal (x, X).
- Hexadecimal digits are broader, including both numbers and letters (a-f, A-F).

The challenge is to ensure your regular expression correctly matches valid cases and rejects invalid cases by following these rules while making it general enough for different formats.

Positive regex:

3.2 Minimization of DFA

When you minimize a DFA, you focus on removing unnecessary transitions and states that don't really add any new information or functionality.

Here's how it works:

- Unnecessary Transitions: These are paths (transitions) that lead to states that are either the same as other states or don't help you make new decisions. If two different states always behave the same way (lead to the same next steps), you can combine them into one state.
- Unnecessary States: If you have two states that do exactly the same thing—meaning no matter where you go from them, the result is always the same—there's no need for both of them. You can remove one of these states and just use the other.

So, the idea is to simplify by combining or removing states and transitions that are redundant, but while still keeping the DFA working the same way as before.

This way, the DFA becomes smaller (with fewer states and transitions), but it still accepts or rejects the same strings (or solves the same problem) as the original one!

3.2.1 Unnecessary Transitions: Transitions Leading to Equivalent States

Sometimes, different states in a DFA behave in the same way for all inputs (as we discussed before). If two states are equivalent (they always transition to the same places and both are either accepting or non-accepting), transitions to these states don't really make a difference because they lead to the same outcome.

In this case, we can merge these equivalent states into a single state. After merging, all transitions that used to go to either state now just go to the merged state.

Example:

- Suppose **State A** and **State B** behave the same (they are equivalent).
- **State C** has a transition to **State A** on input "0" and another transition to **State B** on input "1."

Since **State A** and **State B** behave the same, we don't need two separate transitions anymore. We can just make **State C** have both transitions go to a single, merged state instead, reducing the number of transitions and simplifying the DFA.

Imagine you're in a room with two doors, and both doors lead to the same playground. It doesn't matter which door you pick because you'll end up at the same place.

In a DFA, this means if two states (rooms) always lead to the same result (the playground), we can just keep one door (state) and get rid of the other one. The game still works the same, but now you have one less door to worry about.

3.2.2 Unnecessary Transitions: Transitions Leading to Dead States

A dead state (sometimes called a trap state) is a state where, once you enter it, no matter what input comes next, the DFA can never reach an accepting (final) state. This means that if you go to this state, the DFA will never accept the input string, so it doesn't help in making new decisions.

Transitions that lead to dead states are often not useful, except in specific cases like rejecting input. If you can remove transitions leading to dead states without changing the DFA's ability to reject invalid inputs, you can simplify the DFA.

Example:

- Suppose **State D** is a dead state. Once you reach **State D**, the DFA can never accept any string.
- **State E** has a transition to **State D** on input "0."

If this transition doesn't contribute meaningfully to rejecting strings (for example, if other parts of the DFA can already handle rejections), this transition might be unnecessary and can be removed.

Think of playing a board game. You're on a square that says "You lose" if you land on it, and after that, no matter what you roll, you can't win anymore. That's a dead state.

In a DFA, once you get to a dead state, it's like you've lost, and nothing else matters. Sometimes, the game might have extra ways to lose that don't help, so you can remove those unnecessary paths. As long as the important losing path is still there, the game is simpler and easier to follow.

3.2.3 Unnecessary Transitions: Transitions to Non-Useful States

Some states might not help make new decisions because they either:

Always transition back to themselves, creating a loop that doesn't change the result. Lead to a state that behaves just like another state that's already being used. In both cases, these transitions can be considered unnecessary and might be removed during minimization, making the DFA simpler without affecting its correctness.

Example:

- **State F** has a transition that loops back to itself on every input.
- This loop doesn't help in reaching an accepting state or rejecting anything.

In this case, **State F** might be unnecessary if it doesn't contribute to the overall behavior of the DFA, and you can remove both the state and its transitions.

Imagine you're playing a game where you go from room to room. You find a room that just keeps sending you back to the same room, over and over, with no way out. This room doesn't help you finish the game—it's just a loop.

In a DFA, if a state just keeps looping and doesn't help you win or lose, it's not useful. We can remove that room, and now the game is faster because we don't waste time going in circles.

3.2.4 Unnecessary States: Two states with the same behaviour

A state behaves the same way as another state when, no matter what input you give, both states lead to the same outcome for every possible input. Let me explain this more clearly.

Imagine two states, **State A** and **State B**, in a DFA. These two states behave the same if:

- **Same transitions**: For every possible input (like a letter or number in the alphabet of the DFA), both **State A** and **State B** always move to the same next state. For example, if you give the input "1" to both states, they both move to **State C**, and if you give the input "0" to both states, they both move to **State D**. It doesn't matter where you start; the behavior is identical.
- **Same finality**: Either both states are "accepting" (final) states or both are "non-accepting" (non-final) states. This means if a string ends in **State A** or **State B**, they must either both accept or both reject it.

Example: Imagine you have two toy robots, *Robot A* and *Robot B*. These robots can follow simple instructions: they can either *wave* or *jump* based on what you tell them to do. Your instructions are two buttons:

- **Button 1**: tells the robot to wave.
- **Button 2**: tells the robot to jump.

Now, let's see how both robots behave:

- When you press **Button 1**, both *Robot A* and *Robot B* wave.
- When you press **Button 2**, both *Robot A* and *Robot B* jump.

No matter which button you press, both robots always do the same thing. So, even though you have two robots, you only really need one of them because they behave exactly the same!

In this example:

- *Robot A* and *Robot B* are like the two states in a DFA.
- **Button 1** and **Button 2** are the inputs.

Both robots (states) behave the same way (wave or jump), so we don't need both; we can just keep one robot and simplify things. In a DFA, if two states behave the same way for every possible input, we can combine them into one state, just like keeping one robot that does the same actions.

3.2.5 How to solve DFA minimization task

Task taken from: Neso Academy

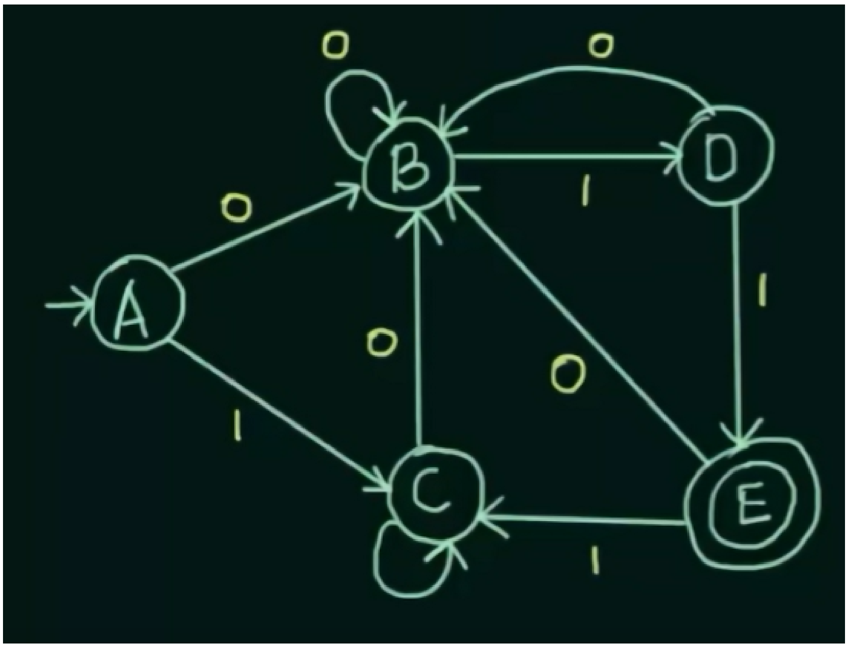


Figure 21: The task to solve

Step 1. Make a state transition table for the DFA.

	0	1
→ A	B	C
B	B	D
C	B	C
D	B	E
E	B	C

Table 1: Caption

Step 2. Find the different equivalences by comparing what states each state transitions to for a given value.

To find 0 equivalence, group all non-final states and final states into separate sets. 0 equivalence: {A,B,C,D} {E}

To determine if two states are 1-equivalent, we compare their transitions on all possible input values and see if they lead to states that were grouped together in the previous equivalence (0-equivalence).

Let's start by comparing State A and State B:

- For input "0", both State A and State B transition to State B. Since they go to the same state, we don't need to look further for this input.

- For input "1", State A transitions to State C, and State B transitions to State D. To determine if State A and State B are equivalent, we check if State C is in the same group as State A in the 0-equivalence step, and if State D is in the same group as State B. Since State C and State D are in the same 0-equivalence set as State A and State B (the non-final set), State A and State B are 1-equivalent and can be placed in the same group.

Next, let's compare State A and State C:

Both State A and State C have the exact same transitions for both inputs, so they are automatically equivalent. Therefore, State A and State C belong in the same group. Now, let's compare State C and State D:

- For input "0", both State C and State D transition to State B. Since they go to the same state, we don't need to check further for this input.
- For input "1", State C transitions to itself, while State D transitions to State E. To check if State C and State D are 1-equivalent, we examine where their transitions lead. State C needs to be in the same 0-equivalence group as itself (the non-final set), while State D needs to transition to a state that is in the same 0-equivalence group as State E. However, since State E is in a different group (the final state set), State C and State D are not equivalent.

Therefore, State C can remain in the same group as State A and State B, but State D must go into its own group.
1 equivalence: {A,B,C} {D} {E}

Now the same steps are repeated but for A, B and C.
2 equivalence: {A,C} {B} {D} {E}

3 equivalence: {A,C} {B} {D} {E}

3.3 Compute the equivalence classes of the set S with respect to equivalence relation

Compute the equivalence classes of the set *S* below with respect to the equivalence relation $=\%_5=$ (same remainder in division by 5).

S = $\{-16, -14, -12, -8, -6, -2, 0, 2, 3, 4, 5, 7, 10, 11, 12, 13, 16, 18, 19, 20\}$

The result must be given as a set of sets.

{ {0,5,10,20}, {-14,11,16}, {-8,2,7,12}, {-12,-2,3,13,18}, {-16,-6,4,19} }

Step 1. Look at the %, which is in this case is 5, write n=5. Set up n sets that start with 0.

- 0:
- 1:
- 2:
- 3:
- 4:

Step 2. Use calculator CASIO → OPTN → Nummerical → MOD

Let's take the first number in the set S, -16 plot it in the MOD along with the equivalence relation %5.

$$MOD(-16, 5) = 4$$

Since the remainder is 4, you place -16 in Set 4, which is actually the 5th set because counting starts from 0.

$$\{\}\{\}\{\}\{\}\{-16\}$$

Continue with each number in the set till you have placed all in their respective sets.

$$\{\{0,5,10,20\},\{-14,11,16\},\{-8,2,7,12\},\{-12,-2,3,13,18\},\{-16,-6,4,19\}\}$$

3.4 How to solve temporal logic

The primary operators include:

- G (Globally): Specifies that a condition must hold for all future states. i.e. Continuous
- F (Finally): Denotes that a condition will be true at some point in the future. i.e. Some
- X (Next): Indicates that a condition must be true in the directly following state.
- U (Until): Represents that a condition must hold true until another condition becomes true.

These operators allow for the expression of complex temporal properties in a concise manner. Consider a safety-critical system where a button press is required to activate an emergency stop. An LTL property could be expressed as:

Example:

G(button_pressed - extgreater F emergency_stop) This formula means, globally, if a button is pressed, it will eventually lead to the activation of the emergency stop.

3.5 How to sort conditions based on strength

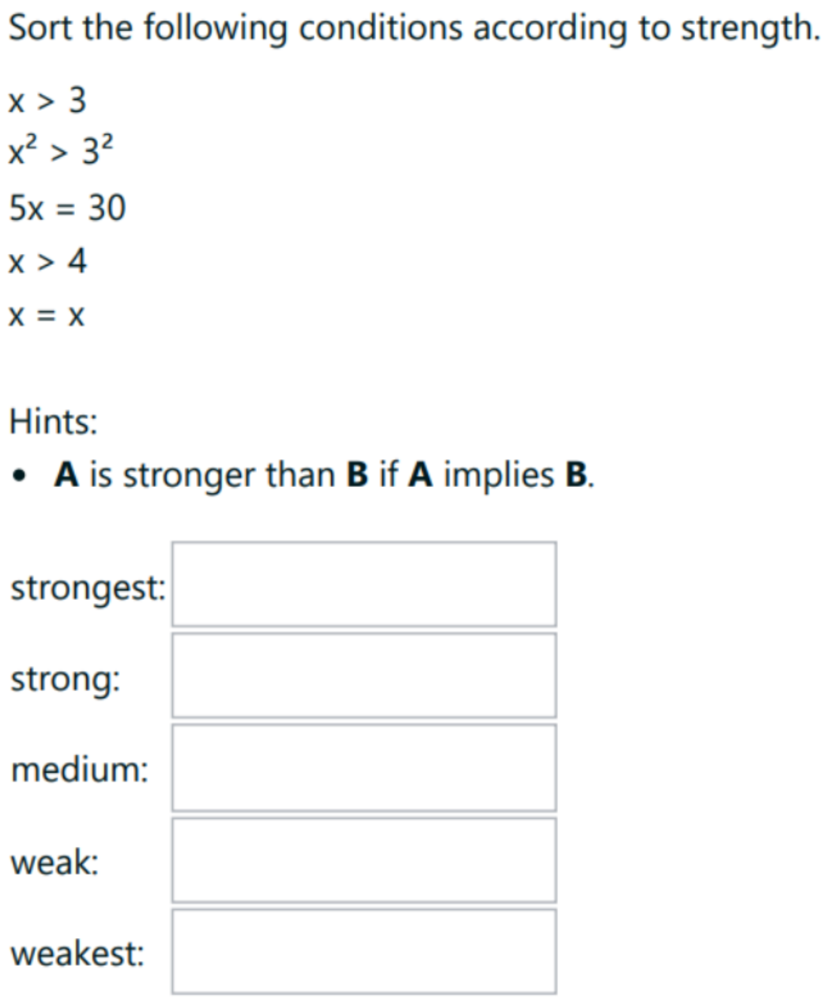


Figure 22: Sort the following conditions according to strength.

Look at each condition and see how many values x can have in total. The less values x can be, the stronger x is. Lets look at $x > 3$, this tells us x can be 4 and above. $x^2 > 3^2$ can be both x $>$ 9 and x $>$ -9, so it has double the amount of $x > 3$ meaning its weaker. $5x = 30$ gives us $x = 6$. Since x is only one value, its the strongest it can be. $x > 4$ tells us x is 5 and above, meaning its stronger than $x > 3$ since it can be 1 value less. $x = x$ means it has infinite amount of possible values, meaning its the weakest. Sorted based on this logic we get the answer:

strongest:	$5x = 30$
strong:	$x > 4$
medium:	$x > 3$
weak:	$x^2 > 3^2$
weakest:	$x = x$

3.6 How to check the properties of a relation R

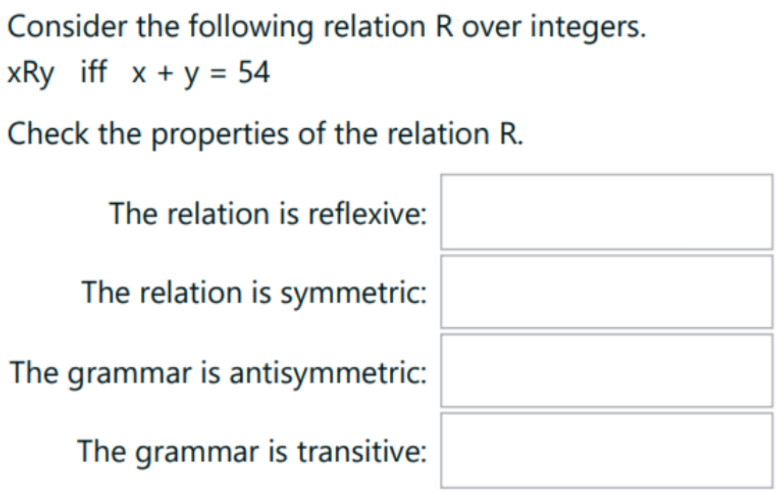


Figure 23: Consider the following relation R over integers. Check the properties of the relation R.

To find out if the relation is reflexive we have to check if $aRa = a + a = x$. This means if $x + x = 54$, meaning

Reflexive: A relation is reflexive if every element is related to itself. Here, for xRx to hold, we would need x+x=54, or 2x=54, which implies x=27. Therefore, only the integer 27 is related to itself. Thus, R is not reflexive over the integers, as this condition does not hold for all integers.

Symmetric: A relation is symmetric if xRy implies yRx. Here, if x+y=54, then y+x=54 as well, which is clearly true. So, the relation is symmetric.

Antisymmetric: A relation is antisymmetric if xRy and yRx imply x=y. Here, if xRy, then x+y=54. However, there are many pairs (x,y) (such as x=20 and y=34) that satisfy xRy and yRx without x=y. Therefore, the relation is not antisymmetric.

Transitive: A relation is transitive if xRy and yRz imply xRz (xRy=yRz=xRZ). For xRy, we have x+y=54, and for yRz, y+z=54. Adding these equations gives x+2y+z=108. For xRz to hold, we would need x+z=54, but this does not necessarily follow from the previous conditions. Therefore, the relation is not transitive.

Based on this logic we get the answer:

reflexive:	false
symmetric:	true
antisymmetric:	false
transitive:	false

3.7 How to construct a regular expression that matches a positive and negative case

Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.
positive cases: hhhhhukk, hhhhhukkk, hhhhukkk, hhhuk, hhuk, hukk, hukkkk, ukk, ukkk, ukkkkk
negative cases: hhhuukkx, hhuuukx, huukkx, huuukkxx, k, u, ukx, uuk, uuukk, uxx

You must fill in all four parts of the answer to be graded.

h+	u?	k+	x

Figure 24: Constructs a regular expressions that matches the following positive cases and does not match the following negative cases.

- **h***:
  - The * symbol means "zero or more of the preceding character."
  - So, h* matches zero or more occurrences of the character 'h'.
  - In this task, it matches the sequence of 'h' characters at the beginning of each positive example.
- **u**:
  - U without any symbols writes just u.
- **k+**:
  - Similar to h+, the k+ means "one or more of the character 'k'."
  - This ensures that there is at least one 'k' in the string, and it can be repeated multiple times, as seen in some positive cases (e.g., hhkkk).

Additional Key Symbols in Regular Expressions:

- . (dot): Represents any single character (except newline characters).
- * (asterisk): Matches zero or more of the preceding character.
- ^: Matches the beginning of a string.
- $: Matches the end of a string.
- [ ]: Matches any one of the characters inside the brackets. For example, [abc] matches either 'a', 'b', or 'c'.
- {m,n}: Matches at least 'm' and at most 'n' occurrences of the preceding character.
- u{3}: Matches exactly three u's.

3.8 How to select all pairs that are in R

Let R be a relation over $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

We know that R is an equivalence relation and a partial order.

Moreover, $(6, 8)$ is not in R.

Select all pairs that are in R.

- [ ] $[1, 4]$
- [ ] $[2, 5]$
- [ ] $[3, 9]$
- [ ] $[4, 6]$
- [ ] $[5, 9]$
- [x] $[6, 6]$
- [ ] $[6, 8]$
- [ ] $[7, 6]$
- [ ] $[7, 9]$
- [ ] $[8, 3]$
- [ ] $[8, 6]$
- [x] $[8, 8]$
- [ ] $[9, 3]$
- [ ] None of the above

Figure 25: Let R be a relation over.

Select all pairs that have equal numbers on both the x and y. In the task above its $[6, 6] and [8, 8]$.

3.9 How to find out which regexes accept the same words as another regex.

Consider the following regular expression.

ex = [A-J]??[0-9]+

Which of the following regular expressions accepts at least the same words as ex?

- [ ] [A-J]?[0-9]?
- [x] [A-J]?[0-9]*
- [x] [A-J]?[0-9]+
- [ ] [A-J]*[0-9]?
- [x] [A-J]*[0-9]*
- [x] [A-J]*[0-9]+
- [ ] [A-J]+[0-9]?
- [ ] [A-J]+[0-9]*
- [ ] [A-J]+[0-9]+
- [ ] None of the above

Figure 26: Consider the following regular expression.

Look at the symbols between the values.

- $? = [0 \to 0.9]$
- $+ = [1 \to \infty]$
- $* = [0 \to \infty]$

If we start by looking at $[A-J]?[0-9]?$. We have a ?, the same as ex, which means we have values from $0 \to 0.9$, so it matches. The next symbol is also a ?, but on ex there is a $+$. $+$ is $[1 \to \infty]$, since ? stops at 0.9 it doesn't contain any other values above 0.9 to infinity, so it doesn't match, therefore $[A-J]?[0-9]?$ is wrong.

Next, we'll look at $[A-J]?[0-9]*$. We know already that ? matches ? in ex. So we'll hop directly to *. * gives values from $[0 \to \infty]$, in ex we have a $+$ which gives values from $[1 \to \infty]$, since both go to infinity * contains all the same values as $+$.

4 Delprøve 3

4.1 Task 1

4.1.1

Find the solution to the initial value problem

$$\begin{cases} a_n - 2 \cdot a_{n-1} - 8 \cdot a_{n-2} = 0 \\ a_0 = -4 \\ a_1 = -10 \end{cases}$$

$a_n =$ -3*(4^n)-(-2)^n

Input your answer as a function of n. Any other variable is treated as a constant. E.g. if you think the answer is $2^n + 3$, you should enter **2^n + 3**.

Figure 27: Find the solution to the initial value problem

4.2 Task 2

4.2.1

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle $1, 2, 3$ in a counter-clockwise manner. Then $r1$ is the rotational symmetry by 120 degrees which sends vertex 1 to vertex 2, and $r2$ is rotation by 240 degrees. The element $t1$ is the reflection symmetry which fixes vertex 1, and likewise for $t2$ and $t3$.

The multiplication table for this group looks like

$$\begin{bmatrix} \text{id} & \text{r1} & \text{r2} & \text{t1} & \text{t2} & \text{t3} \\ \text{r1} & \text{r2} & . & . & \text{t3} & . \\ \text{r2} & \text{id} & \text{r1} & \text{t3} & \text{t1} & \text{t2} \\ \text{t1} & \text{t3} & . & . & \text{r2} & . \\ \text{t2} & \text{t1} & \text{t3} & \text{r1} & \text{id} & \text{r2} \\ \text{t3} & \text{t2} & . & . & \text{r1} & . \end{bmatrix}$$

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3.

$t1 \cdot t1 =$ id
$r1 \cdot t1 =$ t2
$t3 \cdot t1 =$ r2
$t1 \cdot r2 =$ t2
$r1 \cdot r2 =$ id
$t3 \cdot r2 =$ t1
$t1 \cdot t3 =$ r1
$r1 \cdot t3 =$ t1
$t3 \cdot t3 =$ id

Figure 28: Find the entire multiplication table for D_3

4.2.2

Let $(P_7, \bullet_7)$ be the group of natural numbers coprime to 7 with group operation given by multiplication modulo 7. The multiplication table for this group looks like

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & . & 6 & . & 3 & . \\ 3 & . & 2 & . & 1 & . \\ 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & . & 1 & . & 4 & . \\ 6 & 5 & 4 & 3 & 2 & 1 \end{bmatrix}$$

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$3 \bullet_7 2 =$ 6

$2 \bullet_7 2 =$ 4

$5 \bullet_7 2 =$ 3

$3 \bullet_7 4 =$ 5

$2 \bullet_7 4 =$ 1

$5 \bullet_7 4 =$ 6

$3 \bullet_7 6 =$ 4

$2 \bullet_7 6 =$ 5

$5 \bullet_7 6 =$ 2

Figure 29: Compute the missing entries in the group multiplication table

4.3 Task 3

4.3.1

In this problem you are tasked with computing products and inverses in the group $(P_{45}, \bullet_{45})$ of integers and multiplication modulo 45.

$37 \bullet_{45} 41 =$ 32

Your last answer was interpreted as follows:

$$32$$

$41 \bullet_{45} 34 =$ 44

Your last answer was interpreted as follows:

$$44$$

$34 \bullet_{45} 38 =$ 32

Your last answer was interpreted as follows:

$$32$$

$37^{-1} =$ 8

Your last answer was interpreted as follows:

$$8$$

Figure 30: Compute the products and inverses in the group of integers and multiplication

4.3.2

In this problem you are tasked with computing products and inverses in the group $(P_{49}, \bullet_{49})$ of integers and multiplication modulo 49.

$36 \bullet_{49} 27$ = 41

Your last answer was interpreted as follows:

$$41$$

$27 \bullet_{49} 32$ = 31

Your last answer was interpreted as follows:

$$31$$

$32 \bullet_{49} 31$ = 12

Your last answer was interpreted as follows:

$$12$$

36^{-1} = 15

Your last answer was interpreted as follows:

$$15$$

Figure 31: Compute the products and inverses in the group of integers and multiplication

4.4 Task 4

4.4.1

Two players A and B play a game of chance. For each turn, a natural number n in the range 1-100 is chosen at random. If $1 \leq n \leq 46$, player A gives one euro to player B. Otherwise, player B gives one euro to player A. The game ends when one player is broke. The winner is the player which holds all the money in the end.

At the beginning of the game, player A has 44 euros, and player B has 56.

Let p_n be the probability that player A eventually wins, given that she currently has n euros.

If she has $n = 0$ euros, she has lost and the chance of winning is therefore zero. In other words, $p_0 = 0$.

a) Find a 2. order recurrence relation satisfied by p_n:

Figure 32: Find a 2. order recurrence relation satisfied by P_n

b) Compute the probability of player A winning at the moment the game starts:

Figure 33: Compute the probability of player A winning at the moment the game starts

4.5 Task 5

4.5.1

a) Let $G = (\mathbb{Z}/2)^{\times 5}$, and let $H \subset G$ be the subgroup

$$H = \{[0,0,0,0,0], [0,0,0,1,1], [1,0,0,0,0], [1,0,0,1,1]\}.$$

What is the index of $H \subset G$?

$[G : H]$ = 8

Your last answer was interpreted as follows:

$$8$$

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that rotates -180 degrees around the origin.

Compute the index of the subgroup generate by σ:

$[D_4 : \langle \sigma \rangle]$ = 4

Figure 34: Whats the index of $H \subset G$? Compute the index of the subgroup generated by σ.

4.5.2

a) Let $G = (\mathbb{Z}/2)^{\times 4}$, and let $H \subset G$ be the subgroup

$$H = \{[0,0,0,0]\,, [0,0,1,0]\,, [0,1,0,1]\,, [0,1,1,1]\,, [1,0,0,1]\,, [1,0,1,1]\,, [1,1,0,0]\,, [1,1,1,0]\}$$

What is the index of $H \subset G$?

$[G : H]$ = 2

Your last answer was interpreted as follows:

$$2$$

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in I$ symmetry that rotates -90 degrees around the origin.

Compute the index of the subgroup generate by σ:

$[D_4 : \langle\sigma\rangle]$ = 2

Your last answer was interpreted as follows:

$$2$$

Figure 35: Whats the index of $H \subset G$? Compute the index of the subgroup generated by σ.

4.6 Task 6

4.6.1

In this problem you are tasked with computing with elements in the symmetric group of permutations, $(P_5, \cdot)$.

Syntax guide: The answer to each task is an element of the group `$P_5$` and should be given as a 2×5 matrix. E.g. if you think the answer to a question is the permutation denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

Figure 36: Find a 2. order recurrence relation satisfied by P_n

a)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 5 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 4 & 1 \end{bmatrix} =$$

1 2 3 4 5
2 5 3 4 1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 4 & 1 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 2 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 1 & 4 \end{bmatrix} =$$

1 2 3 4 5
2 5 3 4 1

Figure 37: Find a 2. order recurrence relation satisfied by P_n

c)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 5 & 1 & 4 \end{bmatrix}^{-1} =$$

1 2 3 4 5
2 5 3 4 1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 4 & 1 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 4 & 1 \end{bmatrix}^{-1} =$$

1 2 3 4 5
2 5 3 4 1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 4 & 1 \end{bmatrix}$$

Figure 38: Find a 2. order recurrence relation satisfied by P_n

4.7 Task 7

4.7.1

The following sequence

$$p_n = \frac{4^{n+1}}{23}$$

is a special solution to the recurrence relation

$$5 \cdot a_n + 3 \cdot a_{n-1} = 4^n .$$

a) Find another solution to the recurrence relation, not equal to p_n:

$a_n =$ 0

Your last answer was interpreted as follows:

$$0$$

Syntax guide: Your answer should be an expression which depends on the variable n, and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n} ,$$

you should type **-7*(2^n+1) / (3*n)**.

b) Given the initial condition $a_0 = -\frac{19}{23}$. Compute a_n for $n = 1$:

$a_1 =$ 149/115

Your last answer was interpreted as follows:

$$\frac{149}{115}$$

Figure 39: Find another solution to the recurrence relation not equal to P_n. Given the initial condition a_0, compute a_n for $n = 1$

4.7.2

The following sequence

$$p_n = -(-2)^n$$

is a special solution to the recurrence relation

$$-3 \cdot a_n - (-2)^n - 4 \cdot a_{n-1} = 0 .$$

a) Find another solution to the recurrence relation, not equal to p_n:

$a_n =$ -(-2)^n + (-4/3)^n

Your last answer was interpreted as follows:

$$-(-2)^n + \left(\frac{-4}{3}\right)^n$$

The variables found in your answer were: $[n]$

Syntax guide: Your answer should be an expression which depends on the variable n, and *no other constants*. E.g. the answer is

$$-7 \cdot \frac{2^n + 1}{3n} ,$$

you should type **-7*(2^n+1) / (3*n)**.

b) Given the initial condition $a_0 = -3$. Compute a_n for $n = 1$:

$a_1 =$ 14/3

Your last answer was interpreted as follows:

$$\frac{14}{3}$$

Figure 40: Find another solution to the recurrence relation not equal to P_n. Given the initial condition a_0, compute a_n for $n = 1$

4.8 Task 8

4.8.1

In this problem you are tasked with computing subgroups of $(\mathbb{Z}/n, +_n)$ and direct products of such groups.

a) Find the subgroup generated by the element $2 \in \mathbb{Z}/5$.

$< 2 >$ = {0,1,2,3,4}

Your last answer was interpreted as follows:

$$\{0, 1, 2, 3, 4\}$$

Syntax guide: Input your answer as a set. E.g. if you think the answer is $\{1, 2, 3\}$, then you should type **{1,2,3}**.

b) Find the subgroup generated by the element $0 \in \mathbb{Z}/3$.

$< 0 >$ = {0}

Your last answer was interpreted as follows:

$$\{0\}$$

Syntax guide: Input your answer as a set. E.g. if you think the answer is $\{1, 2, 3\}$, then you should type **{1,2,3}**.

c) Find the subgroup generated by the element $[2, 0] \in \mathbb{Z}/5 \times \mathbb{Z}/3$.

$< [2, 0] >$ = {[0,0],[2,0],[4,0],[1,0],[3,0]}

Your last answer was interpreted as follows:

$$\{[0, 0], [2, 0], [4, 0], [1, 0], [3, 0]\}$$

Syntax guide: Elements of this group are on the form $[a, b]$. Your answer should be a set of such elements. E.g. if you think the answer is $\{[1, 2], [0, 2], [1, 3]\}$, then you should type **{[1,2],[0,2],[1,3]}**.

Figure 41: Compute the subgroups of $\mathbb{Z}/n, +_n$ and direct products of such groups

4.9 Task 9

Let A and X be the sets

$$A = \{a, b, c, d\}$$
$$X = \{!, +, -\}.$$

In this problem we consider words written in the alphabet which is the union of the two sets: $A \cup X$. This means that a word in this alphabet can contain letters from both A and X.

However, not all words are valid. The following rules apply:

1. A word must start with a letter from the set A.
2. A word must end with a letter from the set A.
3. A word can not contain two adjacant letters from the set X.

For example the word **a+baa!c** is valid, but the words **a+baa+!c**, **!aaa+c** and **acca!** are not.

The length of a word is the number of letters it consists of. For example, the length of **a+baa!c** is 7.

Let a_n equal the number of valid words of length n.

a) Find a recurrence relation satisfied by a_n:

b) Compute the value

a_1 = 2

Your last answer was interpreted as follows:

$$2$$

c) Compute the value

a_3 = 4

Your last answer was interpreted as follows:

$$4$$

d) Find the general solution to the reccurence relation satisfied by a_n.

a_n = A+n(X+A)

Your last answer was interpreted as follows:

$$A + n(X + A)$$

The variables found in your answer were: $[A, X]$

Figure 42: Find a recurrence relation satisfied by a_n. Compute the value a_n

5 Test 3: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod(n) and Lagrange's theorem

5.1 How to find the solution to the initial value problem

Step 1: Solve the recurrence relation

Find the solution to the initial value problem

$$\begin{cases} a_n - 2 \cdot a_{n-1} - 8 \cdot a_{n-2} = 0 \\ a_0 = -4 \\ a_1 = -10 \end{cases}$$

$a_n =$ -3*(4^n)-(-2)^n

Input your answer as a function of n. Any other variable is treated as a constant. E.g. if you think the answer is $2^n + 3$, you should enter **2^n + 3**.

The recurrence relation is

$$a_n - 2a_{n-1} - 8a_{n-2} = 0$$

This is a second-order linear recurrence relation. The solution to such a relation is always of the form $a_n = C_1 r_1^n + C_2 r_2^n$, where r_1 and r_2 are the roots of the characteristic equation, and C_1 and C_2 are constants determined by the initial conditions. Assume a solution of the form $a_n = r^n$, substitute it into the recurrence relation:

$$r^n - 2r^{n-1} - 8r^{n-2} = 0$$

Divide through by r^{n-2} (for $r \neq 0$):

$$r^n \div r^{n-2} = r^{n-(n-2)} = r^{n-n+2} = r^2$$

$$-2r^{n-1} \div r^{n-2} = -2r^{n-1-(n-2)} = -2r^{n-1-n+2} = -2r^1$$

$$-8r^{n-2} \div r^{n-2} = -8r^{n-2-(n-2)} = -8r^{n-2-n+2} = -8r^0 = -8$$

$$r^2 - 2r - 8 = 0$$

Step 2: Solve the characteristic equation

Solve the quadratic equation $r^2 - 2r - 8 = 0$:

$$\begin{aligned} r &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-8)}}{2(1)} \\ &= \frac{2 \pm \sqrt{4 + 32}}{2} \\ &= \frac{2 \pm 6}{2}. \end{aligned}$$

This gives the roots:

$$r_1 = 4, r_2 = -2$$

Step 3: General Solution The general solution to the recurrence relation is:

$$a_n = C_1(4^n) + C_2((-2)^n)$$

where C_1 and C_2 are constants to be determined using the initial conditions.

Step 4: Apply initial conditions

Using $a_0 = -4$ and $a_1 = -10$:

1. For $n = 0$:
   - $a_0 = C_1(4^0) + C_2((-2)^0) = C_1 + C_2 = -4$
2. For $n = 1$:
   - $a_1 = C_1(4^1) + C_2((-2)^1) = 4C_1 + 2C_2 = -10$

Now we solve the system of equations:

$$\begin{cases} C_1 + C_2 = -4, \\ 4C_1 - 2C_2 = -10. \end{cases}$$

From the first equation:

$$C_2 = -4 - C_1.$$

Substitute $C_2 = -4 - C_1$ into the second equation:

$$4C_1 - 2(-4 - C_1) = -10,$$

$$4C_1 + 8 + 2C_1 = -10,$$

$$6C_1 = -18 \implies C_1 = -3.$$

Substitute $C_1 = -3$ into $C_2 = -4 - C_1$:

$$C_2 = -4 - (-3) = -1.$$

Step 5: Final solution

The solution is:

$$a_n = -3(4^n) - 1((-2)^n).$$

Verify initial conditions

1. For $n = 0$:

$$a_0 = -3(4^0) - 1((-2)^0) = -3 - 1 = -4.$$

2. For $n = 1$:

$$a_1 = -3(4^1) - 1((-2)^1) = -12 + 2 = -10.$$

Both conditions are satisfied.

Final Answer:

$$a_n = -3(4^n) - 1((-2)^n).$$

5.2 How to find the entire multiplication table for D_n

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle $1, 2, 3$ in a counter-clockwise manner. Then $r1$ is the rotational symmetry by 120 degrees which sends vertex 1 to vertex 2, and $r2$ is rotation by 240 degrees. The element $t1$ is the reflection symmetry which fixes vertex 1, and likewise for $t2$ and $t3$.

The multiplication table for this group looks like

$$\begin{bmatrix} \text{id} & \text{r1} & \text{r2} & \text{t1} & \text{t2} & \text{t3} \\ \text{r1} & \text{r2} & . & . & \text{t3} & . \\ \text{r2} & \text{id} & \text{r1} & \text{t3} & \text{t1} & \text{t2} \\ \text{t1} & \text{t3} & . & . & \text{r2} & . \\ \text{t2} & \text{t1} & \text{t3} & \text{r1} & \text{id} & \text{r2} \\ \text{t3} & \text{t2} & . & . & \text{r1} & . \end{bmatrix}$$

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3.

$t1 \cdot t1 =$ id

$r1 \cdot t1 =$ t2

$t3 \cdot t1 =$ r2

$t1 \cdot r2 =$ t2

$r1 \cdot r2 =$ id

$t3 \cdot r2 =$ t1

$t1 \cdot t3 =$ r1

$r1 \cdot t3 =$ t1

$t3 \cdot t3 =$ id

5.3 How compute the missing entries in the group multiplication table

In this problem you are tasked with computing products and inverses in the group $(P_{45}, \bullet_{45})$ of integers and multiplication modulo 45.

$37 \bullet_{45} 41 =$ 32

Your last answer was interpreted as follows:

$$32$$

$41 \bullet_{45} 34 =$ 44

Your last answer was interpreted as follows:

$$44$$

$34 \bullet_{45} 38 =$ 32

Your last answer was interpreted as follows:

$$32$$

$37^{-1} =$ 8

Your last answer was interpreted as follows:

$$8$$

Step 1: Multiply the given numbers

$$2 \cdot 3 = 6$$

Step 2: Take the product and divide it by modulo 7

$$6 \div 7 \approx 0.857$$

We see that the result is less than 1, meaning the modulo is 6. This means if the product is less than the modulo, then the missing entry equals the product.

Lets look at a product that is bigger than the modulo

$$5 \cdot 2 = 10$$

$$10 \div 7 \approx 1.429$$

Step 3: Subtract the modulo from the product

$$10 - 7 = 3$$

Extra: How to do it on calculator

$$OPTN \Rightarrow \triangle \Rightarrow NUMERIC \Rightarrow \triangle \Rightarrow MOD$$

$$MOD(10, 7) = 3$$

5.4 How compute the products and inverses in a group of integers and multiplication modulo

Step 1: Multiply the given numbers

$$37 \cdot 41 = 1517$$

Step 2: Divide the product with modulo 45

$$1517 \div 45 \approx 33.7$$

Step 3: Subtract the answer times the modulo from the product

$$1517 - (33 \cdot 45) = 32$$

The answer is 32.

Extra: How to do it on calculator

$$OPTN \Rightarrow \triangle \Rightarrow NUMERIC \Rightarrow \triangle \Rightarrow MOD$$

$$MOD(1517, 45) = 32$$

For x^{-1}:

Step 1: Make a list in statistics

$$MENU \Rightarrow 2$$

In a free list, in this case List 1, write the numbers from 1 to 50.

Step 2: Use the list in RUN-MATRIX by multiplying it with x

$$MENU \Rightarrow 1$$

$$OPTN \Rightarrow \triangle \Rightarrow NUMERIC \Rightarrow \triangle \Rightarrow MOD$$

$$MOD(List1 \cdot 37, 45)$$

Now we will get a list of the results, look through the answers until you find a 1. In this case at placement 28, we get a 1, so the answer is 28.

$$37^{-1} = 8$$

5.5 How to find the index of $H \subset G$ and compute the index of the subgroup generated by σ

5.5.1

a) Let $G = (\mathbb{Z}/2)^{\times 5}$, and let $H \subset G$ be the subgroup

$$H = \{[0,0,0,0,0], [0,0,0,1,1], [1,0,0,0,0], [1,0,0,1,1]\}.$$

What is the index of $H \subset G$?

$[G : H]$ = 8

Your last answer was interpreted as follows:

8

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that rotates -180 degrees around the origin.

Compute the index of the subgroup generate by σ:

$[D_4 : \langle\sigma\rangle]$ = 4

Task A Step 1: Find out what the total number of elements in G is

$$|G| = 2^5 = 32$$

Step 2: Look at how many elements subgroup H has

$$H = \{[0,0,0,0,0], [0,0,0,1,1], [1,0,0,0,0], [1,0,0,1,1]\}$$

$$|H| = 4 elements$$

Step 3: Find the index of H in G The index $[G : H]$ is the number of cosets of H in G. By the Lagrange theorem, the index is given by:

$$[G : H] = \frac{|G|}{|H|}$$

$$[G : H] = \frac{32}{4} = 8$$

Task B Step 1: Look at the order and the degrees of the rotation If σ rotates -180 8 times, then it makes a circle 4 times, making the answer 4.

6 Delprøve 4

6.1 Task 1

6.1.1

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (1357, 277)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$ 13

Your last answer was interpreted as follows:

$$13$$

It might help to know that the following is a list of the first prime numbers:

$$[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]$$

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$$M = 1228$$

Find the unencrypted (plain text) message

$m =$ 15

Figure 43: Enter Caption

7 Bash

7.1 How to break the RSA encryption

To break Bob's RSA encryption and find his private key, we proceed as follows:

1. RSA Public Key Overview: The RSA public key consists of:

- $n = 2537$: the product of two primes (p and q).
- $e = 1615$: the public exponent.

Bob's RSA key is weak, likely because n is a product of small primes, making factorization feasible.

2. Step 1: Factorize $n = 2537$ To find the private key, we first factor n into its two prime factors, p and q.

We use CASIO: Make a list with 50 numbers

Then do:

$$MOD(2537, List1)$$

Find the number that corresponds with 0, in this case its 43.

Divide n with the found number:

$$2537 \div 43 = 59$$